

are so new that many capital market investors don't play in the same asset pools. Therefore, cryptoassets aren't dancing to the same rhythm of information as traditional capital market assets, at least not yet.

Correlation Coefficient	Effects of Diversification on Risk
+1.0	No risk reduction is possible
+0.5	Moderate risk reduction is possible
0	Considerable risk reduction is possible
-0.5	Most risk can be eliminated
-1.0	All risk can be eliminated

Figure 7.19 ■ The correlation coefficient and effects of diversification on risk

Source: *A Random Walk Down Wall Street*, Burton G. Malkiel, 2015

Figure 7.19 clearly shows that if an asset is zero correlated to other assets in a portfolio, then “considerable risk reduction is possible.” In quantitative terms, reducing risk can be seen by a decrease in the volatility of the portfolio.

If an asset merely reduces the risk of the overall portfolio by being lowly to negatively correlated with other assets, then it doesn't have to provide superior absolute returns to improve the risk-reward ratio of the overall portfolio. Since the Sharpe ratio is returns divided by risk, if the risk gets smaller, then the denominator gets smaller, making the Sharpe ratio bigger. The returns don't have to change at all.

However, it is possible for an asset to be added to a portfolio that both decreases the risk of the portfolio and increases the returns. Finding assets that can do this is rare and